





CS311 final Project

MOViE RENTAL STORE DATABASE

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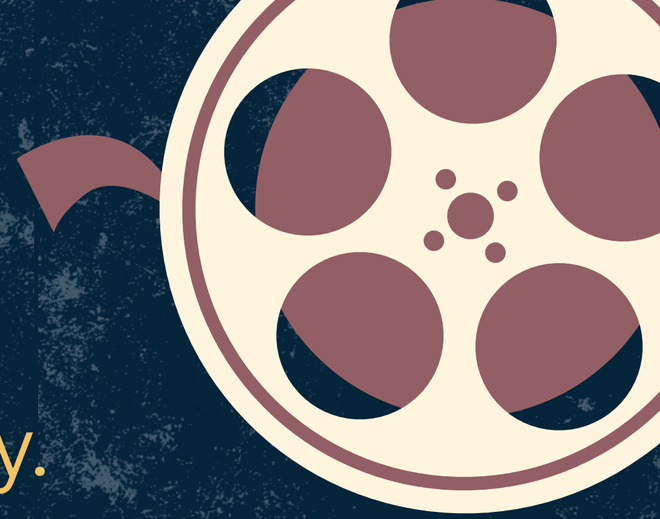
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INTRODUCTION

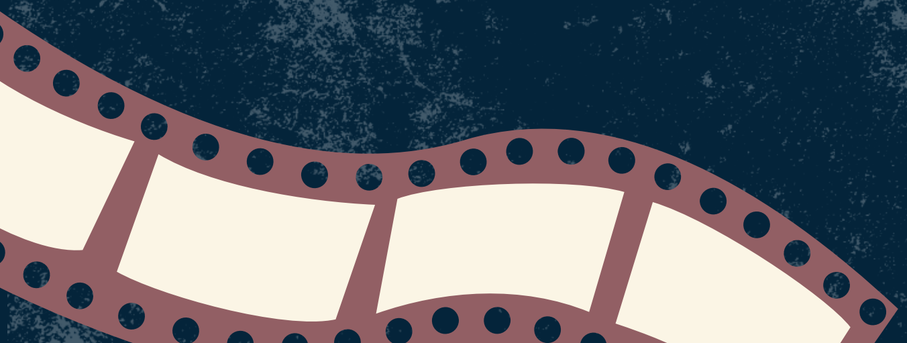
- A Movie Rental Store is a business that allows customers to browse and rent movies for a limited period of time.
- Customers must return the movies before the due date to avoid late fees.
- The store manages important information such as available movies, customers, rental transactions, payments, and staff activities.
- In this project, we designed a database system that supports the daily operations of a Movie Rental Store.
- The system helps organize customer details, track movie rentals and returns, calculate late fees automatically, and maintain accurate and efficient records.
- This database ensures smooth management and improves the overall service provided to customers.



OBJECTiVES OF THE SYSTEM



- To manage customer information efficiently.
- To maintain a complete record of all movies.
- To track rental and return transactions.
- To automate late fee calculations.
- To manage payments processed by staff.



BUSiNESS RULES

1. Customers must be 18 years old or older to create an account.
2. A customer can have up to five active rentals at the same time.
3. Each movie can be rented for a maximum of 10 days.
4. A late fee of 5 SAR per day is applied after the due date.

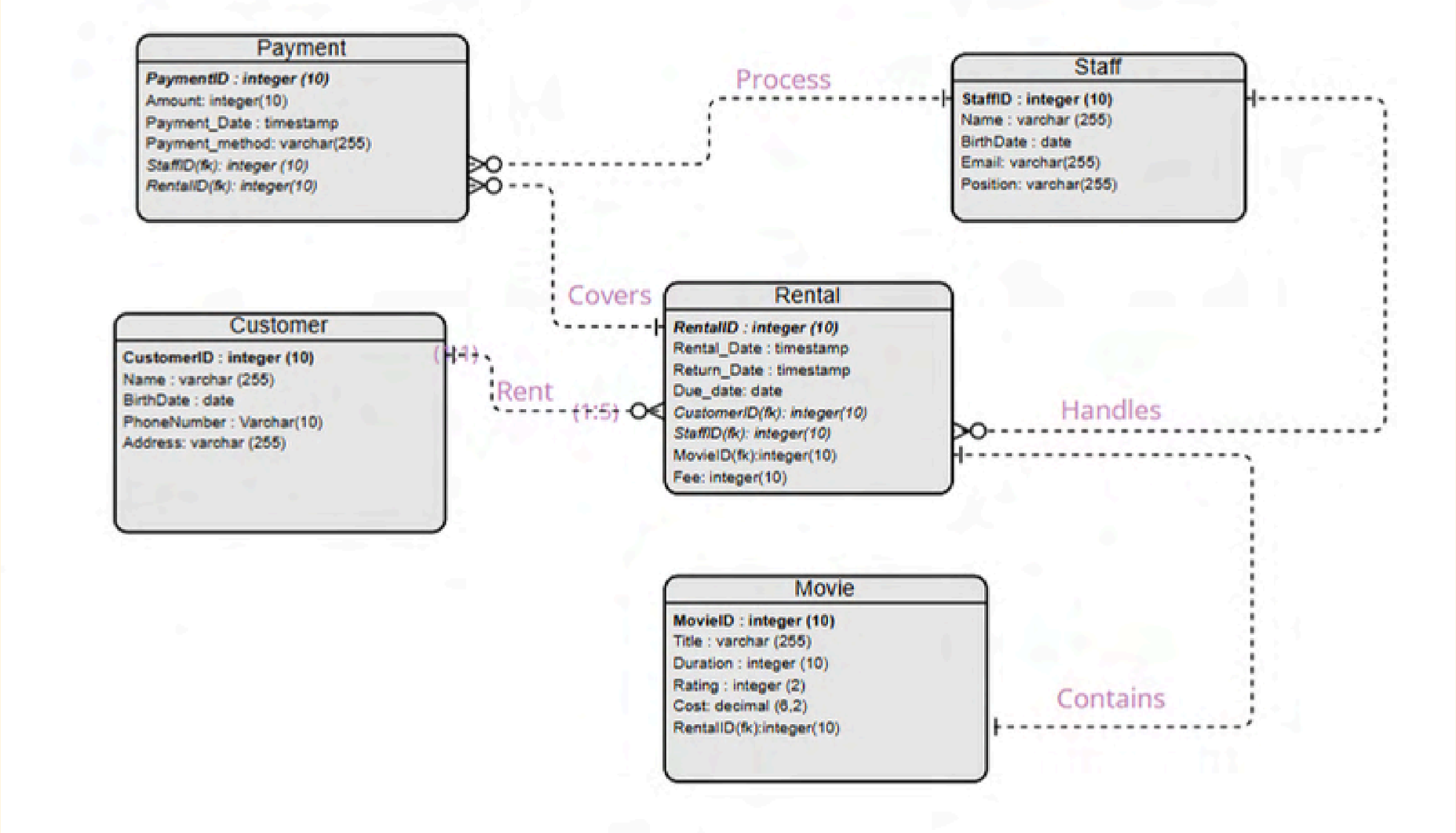
ENTITIES OVERVIEW

The system contains the following main entities

1. CUSTOMER
2. RENTAL
3. MOVIE
4. RENTAL_FEE
5. STAFF
6. PAYMENT

INiTiAL ERD

This is the first draft of the Entity Relationship Diagram:



NORMALiZATiON



STAFF ENTiTY

Staff (Before Normalization)				
StaffID	Name	BirthDate	Email	Position

Staff (After Normalization)					
StaffID (PK)	FirstName	LastName	BirthDate	Email	Position

1NF: PK/ StaffID

Name contains FirstName, LastName

2NF: It is in 1NF, No partial dependencies, it is in 2NF.

3NF: It is in 2NF, No transitive dependencies, it is in 3NF.

CUSTOMER ENTITY

Customer (Before Normalization)				
CustomerID	Name	BirthDate	PhoneNumber	Address

Customer (After Normalization)							
CustomerID (PK)	FirstName	LastName	BirthDate	PhoneNumber	Street	City	Zip_code

1NF: PK/ CustomerID

Name contains FirstName, LastName

Address contains Street, City, Zip_code

2NF: It is in 1NF, No partial dependencies, it is in 2NF.

3NF: It is in 2NF, No transitive dependencies, it is in 3NF.

MOVIE ENTITY

Movie (Before Normalization)				
MovieID	Title	Duration	Rating	Cost

Movie (After Normalization)				
MovieID(PK)	Title	Duration	Rating	Cost

1NF: PK/ MovieID, Atomic, it is in 1NF.

2NF: It is in 1NF, No partial dependencies, it is in 2NF.

3NF: It is in 2NF, No transitive dependencies, it is in 3NF.

RENTAL& RENTAL_FEE ENTITIES

Rental (Before Normalization)						
RentalID	Rental_Date	Return_Date	Due_date	CustomerID	StaffID	Fee

Rental (After Normalization)					
RentalID(PK)	Rental_Date	Return_Date	Due_date	CustomerID	StaffID

Rental_Fee (After Normalization)		
Rental_Fee ID(PK)	RentalID(Fk)	Fee

1NF: PK/ RentalID, Atomic, it is in 1NF.

2NF: It is in 1NF, No partial dependencies, it is in 2NF.

3NF: It is in 2NF, fee depends on RentalID, which is now separated into Rental_Fee No transitive dependencies, it is in 3NF.

PAYMENT ENTITY

Payment (Before Normalization)					
PaymentID	Amount	Payment_date	Payment_method	StaffID	RentalID

Payment (After Normalization)					
PaymentID(PK)	Amount	Payment_date	Payment_method	StaffID (FK)	RentalID(FK)

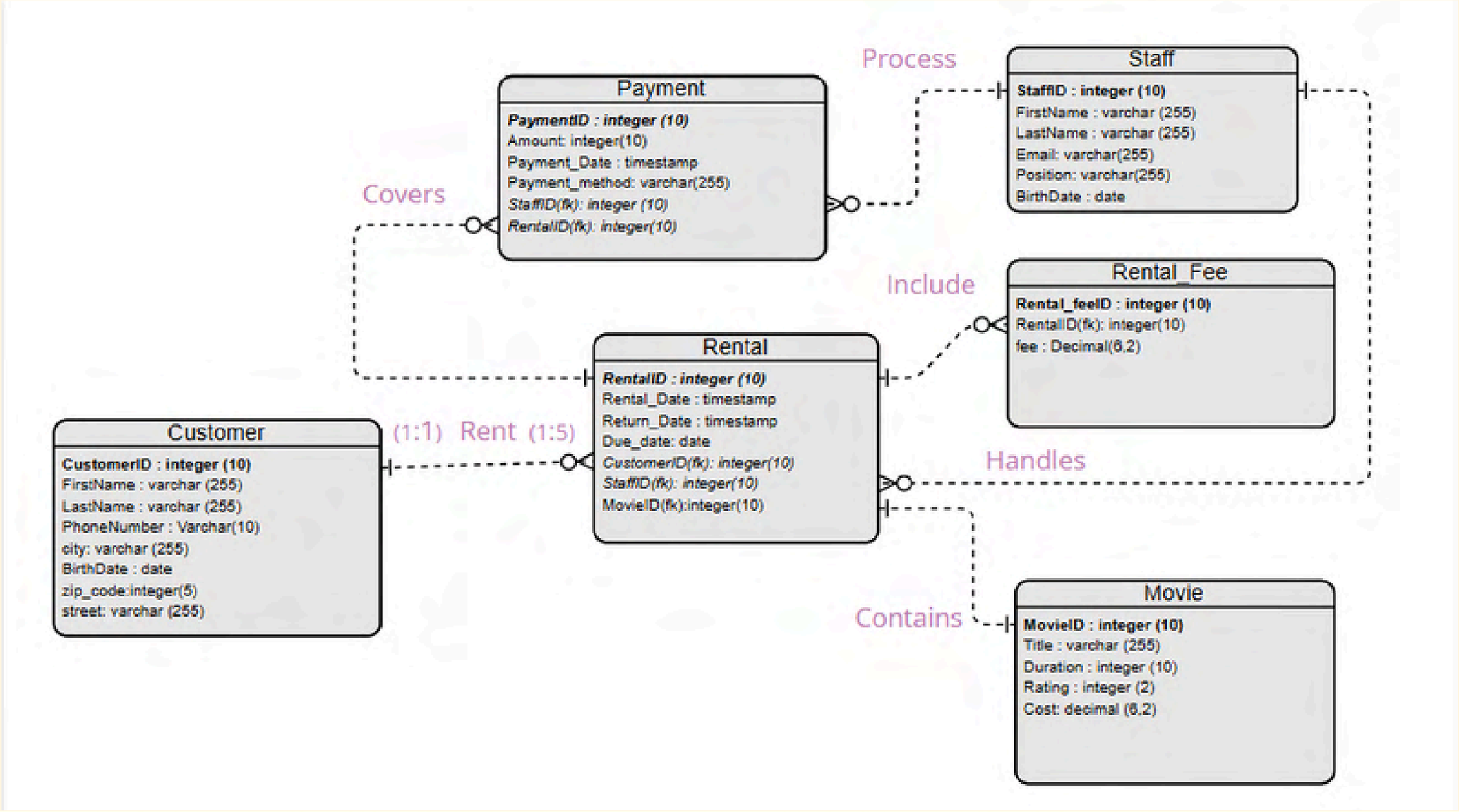
1NF: PK/ PaymentID, Atomic, it is in 1NF.

2NF: It is in 1NF, No partial dependencies, it is in 2NF.

3NF: It is in 2NF, No transitive dependencies, it is in 3NF.

FiNAL ERD

This is the final Entity Relationship Diagram :



DDL



DDL



```sql
create database finalproject;
use finalproject;
create table Customer(
customerID INT primary key  AUTO_INCREMENT NOT NULL,
FirstName VARCHAR (255) NOT NULL,
LastName VARCHAR (255) NOT NULL,
BirthDate Date,
PhoneNumber varchar (10) NOT NULL,
Street VARCHAR (255) NOT NULL,
City VARCHAR (255) NOT NULL,
Zip_cod VARCHAR (15) NOT NULL
);
```

```sql
CREATE TABLE Rental_Fee (
    Rental_FeeID INT PRIMARY KEY AUTO_INCREMENT,
    RentalID INT,
    Fee DECIMAL(10,2) NOT NULL,
    CONSTRAINT FK_Rental FOREIGN KEY (RentalID)
        REFERENCES Rental(RentalID)
        ON DELETE CASCADE
);
```

```sql
CREATE TABLE Movie (
    MovieID     INT PRIMARY KEY AUTO_INCREMENT NOT NULL,
    Title       VARCHAR(255) NOT NULL,
    Duration    INT NOT NULL,
    Rating      INT NOT NULL,
    Cost        DECIMAL(6,2) NOT NULL
);
```

DDL



```sql
CREATE TABLE Rental (
    RentalID INT PRIMARY KEY AUTO_INCREMENT NOT NULL ,
    Rental_Date TIMESTAMP,
    Return_Date TIMESTAMP,
    Due_date DATE,
    CustomerID_FK INT,
    StaffID_FK INT,
    Fee DECIMAL(6,2),

    FOREIGN KEY (CustomerID_FK) REFERENCES Customer(CustomerID),
    FOREIGN KEY (StaffID_FK)    REFERENCES Staff(StaffID)
);
```

```sql
CREATE TABLE Staff (
    StaffID     INT PRIMARY KEY AUTO_INCREMENT NOT NULL
    FirstName   VARCHAR(255) NOT NULL,
    LastName    VARCHAR(255) NOT NULL,
    BirthDate   DATE NOT NULL,
    Email       VARCHAR(255) NOT NULL,
    Position    VARCHAR(255) NOT NULL
);
```

```sql
CREATE TABLE Payment (
    PaymentID INT PRIMARY KEY AUTO_INCREMENT NOT NULL,
    Amount DECIMAL(10,2),
    Payment_Date TIMESTAMP,
    Payment_method VARCHAR(100),
    StaffID_FK INT,
    RentalID_FK INT,
    FOREIGN KEY (StaffID_FK) REFERENCES Staff(StaffID),
    FOREIGN KEY (RentalID_FK) REFERENCES Rental(RentalID)
);
```

DML



DML



```
72
73 •  INSERT INTO Customer (FirstName, LastName, BirthDate, PhoneNumber, Street, City, Zip_cod)
74    VALUES
75    ('Sara','Alharbi','2000-05-10',551234567,'King Street','Riyadh',11564),
76    ('Laila','Almalki','1998-11-20',559876543,'Prince Street','Dammam',31456),
77    ('Shahad','Alotaibi','1995-02-14',568882222,'Corniche','Jeddah',23456),
78    ('Noor','Alqahtani','2000-08-22',557777888,'Olaya','Riyadh',11565),
79    ('Dana','Almutairi','1997-12-05',556666555,'Al-Faisaliah','Riyadh',11566),
80    ('Wejdan','Alghamdi','1999-03-15',554444333,'Prince Street','Dammam',31457),
81    ('Noura','Alshammari','1998-09-10',553333222,'Jeddah Road','Jeddah',23457),
82    ('Fatimah','Alshehri','1996-06-25',552222111,'King Street','Riyadh',11567),
83    ('Jory','Alzahrani','2001-07-12',551111000,'Corniche','Jeddah',23458),
84    ('Alya','Alenzi','2000-01-30',559999888,'Prince Street','Dammam',31458);
85
```

Output

Action Output

#	Time	Action
4	12:46:02	CREATE TABLE Movie (MovieID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, Title VARCHAR(2...
5	12:46:02	CREATE TABLE Staff (StaffID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, FirstName VARCHAR(2...
6	12:46:02	CREATE TABLE Rental (RentalID INT PRIMARY KEY AUTO_INCREMENT NOT NULL , Rental_Date TIMESTA...
7	12:46:02	CREATE TABLE Rental_Details (Rental_detailID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, MovieID...
8	12:46:02	CREATE TABLE Payment (PaymentID INT PRIMARY KEY AUTO_INCREMENT NOT NULL, Amount DECIMAL(...
9	12:46:03	INSERT INTO Customer (FirstName, LastName, BirthDate, PhoneNumber, Street, City, Zip_cod) VALUES ('Sara','Alharb...

DML

```
1 •   SELECT * FROM finalproject.customer;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Con

customerID	FirstName	LastName	BirthDate	PhoneNumber	Street	City	Zip_cod
1	Sara	Alharbi	2000-05-10	551234567	King Street	Riyadh	11564
2	Laila	Almalki	1998-11-20	559876543	Prince Street	Dammam	31456
3	Shahad	Alotaibi	1995-02-14	568882222	Corniche	Jeddah	23456
4	Noor	Alqahtani	2000-08-22	557777888	Olaya	Riyadh	11565
5	Dana	Almutairi	1997-12-05	556666555	Al-Faisaliah	Riyadh	11566
6	Wejdan	Alghamdi	1999-03-15	554444333	Prince Street	Dammam	31457
7	Noura	Alshammari	1998-09-10	553333222	Jeddah Road	Jeddah	23457

DML

```
86 •    INSERT INTO Movie (Title, Duration, Rating, Cost)
87      VALUES
88      ('The Matrix',136,5,30.00),
89      ('Inception',148,5,35.00),
90      ('Titanic',195,4,25.00),
91      ('The Lion King',90,5,20.90),
92      ('Avengers: Endgame',181,5,40.00),
93      ('Joker',122,4,30.90),
94      ('Frozen',102,4,25.00),
95      ('Parasite',132,5,35.00),
96      ('Interstellar',169,5,40.00),
97      ('The Godfather',175,5,50.00);
98
```

Output

Action Output

#	Time	Action
7	12:46:02	CREATE TABLE Rental_Details (Rental_detailID INT PRIMARY KEY
8	12:46:02	CREATE TABLE Payment (PaymentID INT PRIMARY KEY AUTO_IN
9	12:46:03	INSERT INTO Customer (FirstName, LastName, BirthDate, PhoneNumbe

```
1 •    SELECT * FROM finalproject.movie;
```

Result Grid | Filter Rows: | Edit:

MovieID	Title	Duration	Rating	Cost
1	The Matrix	136	5	30.00
2	Inception	148	5	35.00
3	Titanic	195	4	25.00
4	The Lion King	90	5	20.90
5	Avengers: Endgame	181	5	40.00
6	Joker	122	4	30.90
7	Frozen	102	4	25.00

movie 1

DML

```sql
INSERT INTO Staff (FirstName, LastName, BirthDate, Email, Position)
VALUES
('Ahmed','Alqahtani','1985-05-12','ahmed.staff@example.com','Manager'),
('Sara','Alharbi','1990-08-20','sara.staff@example.com','Cashier'),
('Omar','Almutairi','1988-03-15','omar.staff@example.com','Clerk'),
('Laila','Alshehri','1992-11-10','laila.staff@example.com','Cashier'),
```

```sql
);
INSERT INTO RentalFee (RentalID, Fee) VALUES (101, 25.50);
INSERT INTO RentalFee (RentalID, Fee) VALUES (102, 15.00);
INSERT INTO RentalFee (RentalID, Fee) VALUES (103, 30.75);
INSERT INTO RentalFee (RentalID, Fee) VALUES (104, 10.00);
INSERT INTO RentalFee (RentalID, Fee) VALUES (105, 22.25);
```

```sql
SELECT * FROM finalproject.staff;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

StaffID	FirstName	LastName	BirthDate	Email	Position
1	Ahmed	Alqahtani	1985-05-12	ahmed.staff@example.com	Manager
2	Sara	Alharbi	1990-08-20	sara.staff@example.com	Cashier
3	Omar	Almutairi	1988-03-15	omar.staff@example.com	Clerk
4	Laila	Alshehri	1992-11-10	laila.staff@example.com	Cashier
5	Nasser	Alghamdi	1987-07-25	nasser.staff@example.com	Clerk
6	Mona	Alzahrani	1991-09-05	mona.staff@example.com	Manager

DML



```sql
INSERT INTO Rental (Rental_Date, Return_Date, Due_date, CustomerID_FK, StaffID_FK, Fee)
VALUES
('2025-11-01','2025-11-05','2025-11-04',1,3,5.00),
('2025-11-02','2025-11-08','2025-11-07',2,5,5.00),
('2025-11-03','2025-11-07','2025-11-07',3,2,0.00),
('2025-11-04','2025-11-10','2025-11-09',4,1,5.00),
```

```sql
INSERT INTO Payment (Amount, Payment_Date, Payment_method, StaffID_FK, RentalID_FK)
VALUES
(44.9 ,'2025-11-05','Cash',3,1),
(32.7 ,'2025-11-08','Card',5,2),
(58.4 ,'2025-11-07','Cash',2,3),
```

DML

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell C

PaymentID	Amount	Payment_Date	Payment_method	StaffID_FK	RentalID_FK
1	44.90	2025-11-05 00:00:00	Cash	3	1
2	32.70	2025-11-08 00:00:00	Card	5	2
3	58.40	2025-11-07 00:00:00	Cash	2	3
4	44.90	2025-11-10 00:00:00	Card	1	4
5	29.50	2025-11-08 00:00:00	Cash	4	5
6	47.20	2025-11-12 00:00:00	Card	6	6

sult Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell

RentalID	Rental_Date	Return_Date	Due_date	CustomerID_FK	StaffID_FK	Fee
1	2025-11-01 00:00:00	2025-11-05 00:00:00	2025-11-04	1	3	5.00
2	2025-11-02 00:00:00	2025-11-08 00:00:00	2025-11-07	2	5	5.00
3	2025-11-03 00:00:00	2025-11-07 00:00:00	2025-11-07	3	2	0.00
4	2025-11-04 00:00:00	2025-11-10 00:00:00	2025-11-09	4	1	5.00
5	2025-11-05 00:00:00	2025-11-08 00:00:00	2025-11-08	5	4	0.00
6	2025-11-06 00:00:00	2025-11-12 00:00:00	2025-11-12	6	6	0.00

TRiGGERS



TRiGGERS



```
DELIMITER $$
-- Trigger to check customer>18 age
CREATE TRIGGER CheckCustomerAge
BEFORE INSERT ON Customer
FOR EACH ROW
BEGIN
    -- Calcult DateOfBirth
    IF TIMESTAMPDIFF(YEAR, NEW.BirthDate, CURDATE()) < 18 THEN
        -- Raise an error if the Customer is younger than 18
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Customer must be at least 18 years old to be added to the system.';
    END IF;
END$$
DELIMITER ;
```

TRiGGERS



```sql
use finalproject;
INSERT INTO Customer (FirstName, LastName, BirthDate, PhoneNumber, Street, City, Zip_cod)
VALUES ('Mona', 'Ali', '2010-05-10', '0555555555', 'King St', 'Riyadh', '11564');
```

Action Output

	Time	Action	Message
1	22:24:08	use finalproject	0 row(s) affected
2	22:24:08	INSERT INTO Customer (FirstName, LastName, BirthDate, PhoneNumber, Street, City, Zip_cod) VALUES ('Mona', 'Ali',...	Error Code: 1644. Customer must be at least 18 years old to be added to the system.

TRiGGERS



```
DELIMITER $$
CREATE TRIGGER trgmax_rentals BEFORE INSERT ON Rental
FOR EACH ROW
BEGIN
    IF (SELECT COUNT(*)
        FROM Rental
        WHERE CustomerID_FK = NEW.CustomerID_FK and Return_Date IS NULL) >= 5 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Customer cannot have more than 5 active rentals';
    END IF;
END$$
//
DELIMITER ;
```

TRiGGERS



```sql
use finalproject;
INSERT INTO Rental (Rental_Date, Return_Date, Due_date, CustomerID_FK, StaffID_FK)
VALUES ('2025-12-01', NULL, '2025-12-05', 1, 1);

```

Output

Action Output

#	Time	Action	Message
1	22:26:49	use finalproject	0 row(s) affected
2	22:26:49	INSERT INTO Rental (Rental_Date, Return_Date, Due_date, CustomerID_FK, StaffID_FK) VALUES ('2025-12-01', N...	Error Code: 1644. Customer cannot have more than 5 active rentals

TRiGGERS



```
DELIMITER $$
CREATE TRIGGER trg_late_fee
BEFORE INSERT ON Rental
FOR EACH ROW
BEGIN
    IF NEW.Return_Date > NEW.Due_date THEN
        SET NEW.Fee = DATEDIFF(NEW.Return_Date, NEW.Due_date) * 5;
    ELSE
        SET NEW.Fee = 0;
    END IF;
END$$
DELIMITER ;
```

TRiGGERS



```
use finalproject;
INSERT INTO Rental (Rental_Date, Return_Date, Due_date, CustomerID_FK, StaffID_FK)
VALUES ('2025-12-01', '2025-12-14', '2025-12-11', 3, 1);
```

RentalID	Rental_Date	Return_Date	Due_date	CustomerID_FK	StaffID_FK	Fee
18	2025-12-01 00:00:00	2025-12-14 00:00:00	2025-12-11	3	1	15.00

TRiGGERS



```
DELIMITER $$
CREATE TRIGGER trgmax_movie_days
BEFORE INSERT ON Rental
FOR EACH ROW
BEGIN
    IF DATEDIFF(NEW.Due_date, NEW.Rental_Date) > 10 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Movie cannot be rented for more than 10 days';
    END IF;
END$$
DELIMITER ;
```

TRiGGERS



```
1 • use finalproject;
2 • INSERT INTO Rental (Rental_Date, Return_Date, Due_date, CustomerID_FK, StaffID_FK)
3   VALUES ('2025-12-01', NULL, '2025-12-20', 2, 1);
4
```

Output

Action Output

#	Time	Action	Message
1	22:27:56	use finalproject	0 row(s) affected
2	22:27:56	INSERT INTO Rental (Rental_Date, Return_Date, Due_date, CustomerID_FK, StaffID_FK) VALUES ('2025-12-01', N...	Error Code: 1644. Movie cannot be rented for more than 10 days

TRiGGERS



Scripting Help

SQL File 3* SQL File 4*

Limit to 1000 rows

```sql
use finalproject;
DELIMITER $$

CREATE TRIGGER limit_rentals_per_customer
BEFORE INSERT ON Rental
FOR EACH ROW
BEGIN
    IF (SELECT COUNT(*) FROM Rental WHERE CustomerID = NEW.CustomerID) >= 5 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Each customer can have a maximum of 5 rentals.';
    END IF;
END$$

DELIMITER ;
```

STATEMENT



STATEMENT

```sql
use finalproject;
SELECT FirstName, LastName, City, BirthDate
FROM Customer
WHERE (City = 'Riyadh' OR City = 'Dammam')
  AND BirthDate < '2000-01-01';
```

Result Grid | Filter Rows: | Export:

FirstName	LastName	City	BirthDate
Laila	Almalki	Dammam	1998-11-20
Dana	Almutairi	Riyadh	1997-12-05
Wejdan	Alghamdi	Dammam	1999-03-15
Fatimah	Alshehri	Riyadh	1996-06-25

```sql
use finalproject;
SELECT c.FirstName, c.LastName, m.Title, rd.Description
FROM Rental r
JOIN Customer c ON r.CustomerID_FK = c.CustomerID
JOIN Rental_Details rd ON r.RentalID = rd.RentalID_FK
JOIN Movie m ON rd.MovieID_FK = m.MovieID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

FirstName	LastName	Title	Description
Sara	Alharbi	The Matrix	Rented Matrix for 4 days
Sara	Alharbi	Inception	Rented Inception for 4 days
Laila	Almalki	Titanic	Rented Titanic for 6 days
Shahad	Alotaibi	The Lion King	Rented Lion King for 4 days
Noor	Alqahtani	Avengers: Endgame	Rented Avengers for 6 days
Dana	Almutairi	Joker	Rented Joker for 3 days
Wejdan	Alghamdi	Frozen	Rented Frozen for 6 days

STATEMENT



```sql
use finalproject;
SELECT c.FirstName, c.LastName, SUM(p.Amount) AS TotalPayments
FROM Payment p
JOIN Rental r ON p.RentalID_FK = r.RentalID
JOIN Customer c ON r.CustomerID_FK = c.CustomerID
GROUP BY c.FirstName, c.LastName;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

FirstName	LastName	TotalPayments
Laila	Almalki	32.70
Shahad	Alotaibi	58.40
Noor	Alqahtani	44.90
Dana	Almutairi	29.50
Wejdan	Alghamdi	47.20
Noura	Alshammari	61.90
Fatimah	Alshehri	33.80
Jory	Alzahrani	44.90

```sql
use finalproject;
SELECT CustomerID, FirstName, LastName, BirthDate,
       TIMESTAMPDIFF(YEAR, BirthDate, CURDATE()) AS Age
FROM Customer
WHERE TIMESTAMPDIFF(YEAR, BirthDate, CURDATE()) >= 18;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CustomerID	FirstName	LastName	BirthDate	Age
1	Sara	Alharbi	2000-05-10	25
2	Laila	Almalki	1998-11-20	27
3	Shahad	Alotaibi	1995-02-14	30
4	Noor	Alqahtani	2000-08-22	25
5	Dana	Almutairi	1997-12-05	28
6	Wejdan	Alghamdi	1999-03-15	26

THANK YOU

